

Assignment 1 – CS201

Proofs by Contrapositive

Solution 1.

Suppose x is even or y is even.

Case I. If x is even, then x^2 is even, so $x^2(y^2 - 2y)$ is even.

Case II. If y is even, let $y = 2k$. Then $y^2 - 2y = 4k(k - 1)$, which is even. Thus the product is even.

Hence if $x^2(y^2 - 2y)$ is odd, both x, y **must be odd**.

Solution 2. Suppose $a < 8$ and $b < 8$. Then $a \leq 7, b \leq 7$ and $a + b \leq 14 < 15$. Hence, if $a + b \geq 15$, at least one of $a \geq 8$ or $b \geq 8$ must hold.

Proofs by Contradiction

Solution 1. Assume $\sqrt{3}$ is rational: $\sqrt{3} = \frac{p}{q}$ in lowest terms. Then $p^2 = 3q^2$, so p^2 divisible by 3, hence p divisible by 3. Write $p = 3k$, then $9k^2 = 3q^2 \implies q^2 = 3k^2$, so q divisible by 3. Then p, q not coprime, contradiction. Hence $\sqrt{3}$ irrational.

Solution 2. Assume $2\sqrt{2}$ is rational: $2\sqrt{2} = \frac{p}{q} \implies \sqrt{2} = \frac{p}{2q}$ rational. Contradiction since $\sqrt{2}$ irrational. Hence $2\sqrt{2}$ irrational.

Solution 3. Assume $\sqrt{2} + \sqrt{6}$ rational. Then $\sqrt{6} = r - \sqrt{2}$ for some rational r . Squaring: $6 = r^2 + 2 - 2r\sqrt{2} \implies \sqrt{2} = \frac{r^2 - 4}{2r}$. RHS rational, so $\sqrt{2}$ rational, contradiction. Thus $\sqrt{2} + \sqrt{6}$ irrational.

Solution 4. Assume $2^{1/3}$ rational: $2^{1/3} = \frac{p}{q}$ with $\gcd(p, q) = 1$. Cubing: $2q^3 = p^3$, so p^3 even, p even $\implies p = 2k$. Then $2q^3 = 8k^3 \implies q^3 = 4k^3$, so q even. Contradiction. Hence $2^{1/3}$ irrational.

Solution 5.

Proof. Suppose, for contradiction, that there exists an integer $n \geq 2$ that has two distinct prime factorizations. Choose the smallest such n , and write the two (assumed different) factorizations as

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where each p_i and each q_j is prime. By “distinct” we mean the multisets $\{p_1, \dots, p_r\}$ and $\{q_1, \dots, q_s\}$ are not identical.

We need the following standard lemma.

Lemma (Euclid’s lemma). If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof of Lemma. If $p \mid a$ we are done. Otherwise $\gcd(p, a) = 1$. By Bézout’s identity there exist integers x, y with $xp + ya = 1$. Multiplying by b gives

$$xpb + yab = b.$$

The left-hand side is divisible by p , so $p \mid b$. This proves the lemma. $\square$

Apply Euclid’s lemma to the prime p_1 and the product $q_1 \cdots q_s$. Since $p_1 \mid n = q_1 \cdots q_s$, the lemma implies p_1 divides some q_j . But q_j is prime, so $p_1 = q_j$.

Cancel the common prime p_1 from both sides to obtain

$$\frac{n}{p_1} = p_2 \cdots p_r = q_1 \cdots q_{j-1} q_{j+1} \cdots q_s.$$

The two factorizations of n/p_1 are still (by construction) distinct; hence n/p_1 is a smaller integer than n that admits two distinct prime factorizations. This contradicts the minimality of n .

Therefore no such n exists, and every integer $n \geq 2$ has a unique prime factorization (up to the order of the factors). $\square$

Proofs by Mathematical Induction

Solution 1. (*Sum of first n odd numbers equals n^2 .*)

Proof. We prove by ordinary induction on $n \geq 1$.

Base case ($n = 1$). The sum of the first odd number is 1, and $1^2 = 1$, so the statement holds.

Inductive step. Assume for some $n \geq 1$ that

$$\sum_{k=1}^n (2k-1) = n^2.$$

Then for $n+1$,

$$\sum_{k=1}^{n+1} (2k-1) = \left(\sum_{k=1}^n (2k-1) \right) + (2(n+1)-1) = n^2 + (2n+1) = (n+1)^2.$$

Thus the statement holds for $n + 1$. By induction the formula holds for all $n \geq 1$. $\square$

Solution 2. (*Formula for sum of squares:* $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.)

Proof. We prove by induction on $n \geq 1$.

Base case ($n = 1$). Left side $= 1^2 = 1$, right side $\frac{1 \cdot 2 \cdot 3}{6} = 1$.

True.

Inductive step. Assume for some $n \geq 1$ that

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Then

$$\sum_{k=1}^{n+1} k^2 = \left(\sum_{k=1}^n k^2 \right) + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2.$$

Factor $(n+1)$:

$$\frac{n(2n+1)}{6}(n+1) + (n+1)^2 = (n+1) \left(\frac{n(2n+1)}{6} + (n+1) \right).$$

Put over common denominator 6:

$$(n+1) \left(\frac{n(2n+1) + 6(n+1)}{6} \right) = (n+1) \left(\frac{2n^2 + n + 6n + 6}{6} \right) = (n+1) \left(\frac{2n^2 + 7n + 6}{6} \right).$$

Factor the quadratic $2n^2 + 7n + 6 = (2n+3)(n+2)$ to obtain

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6},$$

which is the desired formula with n replaced by $n + 1$. Thus the formula holds for all $n \geq 1$. $\square$

Solution 3. *Claim:* For every integer $n \geq 2$, $n^3 - n$ is divisible by 6.

We prove by induction on n .

Base case ($n = 2$).

$$2^3 - 2 = 8 - 2 = 6,$$

which is divisible by 6.

Inductive step. Assume for some $k \geq 2$ that $k^3 - k$ is divisible by 6; i.e., $k^3 - k = 6t$ for some integer t . Consider $k + 1$:

$$(k+1)^3 - (k+1) = (k^3 + 3k^2 + 3k + 1) - k - 1 = (k^3 - k) + 3k(k+1).$$

By the inductive hypothesis, $(k^3 - k)$ is a multiple of 6. Also, among k and $k + 1$ one is even, so $k(k + 1)$ is even; hence $3k(k + 1)$ is a multiple of 6. Therefore $(k + 1)^3 - (k + 1)$ is a sum of two multiples of 6, hence itself a multiple of 6.

By induction, $n^3 - n$ is divisible by 6 for all $n \geq 2$.

(Remark for intuition.) Note $n^3 - n = n(n - 1)(n + 1)$ is the product of three consecutive integers, which is always divisible by 2 and by 3, hence by 6.

Solution 4. (*Number of subsets of $\{1, 2, \dots, n\}$ is 2^n .*)

Proof. We prove by induction on $n \geq 1$.

Base case ($n = 1$). The set $\{1\}$ has two subsets: $\emptyset$ and $\{1\}$, and $2^1 = 2$.

Inductive step. Assume every n -element set has exactly 2^n subsets. Consider a set of size $n + 1$, say $S_{n+1} = \{1, 2, \dots, n, n + 1\}$. Every subset $T \subseteq S_{n+1}$ either *does not contain* the element $n + 1$ or *does contain* $n + 1$. The subsets that do not contain $n + 1$ are precisely the subsets of $\{1, \dots, n\}$ (there are 2^n of them by the inductive hypothesis). The subsets that do contain $n + 1$ are in bijection with subsets of $\{1, \dots, n\}$ (send $U \subseteq \{1, \dots, n\}$ to $U \cup \{n + 1\}$), so there are another 2^n of them. Therefore the total number of subsets of S_{n+1} is $2^n + 2^n = 2^{n+1}$. This completes the induction. $\square$

Solution 5. (*$11^n - 6$ is divisible by 5 for all $n \in \mathbb{N}$.*)

Proof. Two short proofs are given; the first uses induction (as requested), the second is a quick modular argument.

(Induction proof.) Let $P(n)$ be the statement " $11^n - 6$ is divisible by 5."

Base case ($n = 1$). $11^1 - 6 = 5$, which is divisible by 5.

Inductive step. Assume $11^n - 6$ is divisible by 5 for some $n \geq 1$, i.e. $11^n - 6 = 5k$ for some integer k . Then

$$11^{n+1} - 6 = 11 \cdot 11^n - 6 = 11(11^n - 6) + (11 \cdot 6 - 6) = 11(11^n - 6) + 60.$$

By the inductive hypothesis $11(11^n - 6)$ is divisible by 5, and 60 is divisible by 5, so their sum $11^{n+1} - 6$ is divisible by 5. Hence $P(n+1)$ holds. By induction $11^n - 6$ is divisible by 5 for all $n \geq 1$.

(*Modular proof.*) Observe $11 \equiv 1 \pmod{5}$, so $11^n \equiv 1^n \equiv 1 \pmod{5}$. Therefore $11^n - 6 \equiv 1 - 6 \equiv -5 \equiv 0 \pmod{5}$, i.e. $11^n - 6$ is divisible by 5 for every $n \geq 1$. $\square$

Solution 6. (Recurrence $a_1 = 1, a_2 = 2, a_3 = 3, a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 4$; prove $a_n < 2^n$ for all $n \in \mathbb{N}$.)

Proof. We use *strong induction* (because the recurrence uses three previous terms).

Base cases. Check $n = 1, 2, 3$:

$$a_1 = 1 < 2^1 = 2, \quad a_2 = 2 < 2^2 = 4, \quad a_3 = 3 < 2^3 = 8.$$

So the statement holds for $n = 1, 2, 3$.

Inductive hypothesis. Suppose for some $m \geq 3$ that for all k with $1 \leq k \leq m$ we have $a_k < 2^k$.

Inductive step. Consider a_{m+1} . By the recurrence,

$$a_{m+1} = a_m + a_{m-1} + a_{m-2}.$$

Applying the inductive hypothesis to each term on the right-hand side,

$$a_{m+1} < 2^m + 2^{m-1} + 2^{m-2}.$$

Factor 2^{m-2} :

$$2^m + 2^{m-1} + 2^{m-2} = 2^{m-2}(4 + 2 + 1) = 7 \cdot 2^{m-2}.$$

Since $7 \cdot 2^{m-2} < 8 \cdot 2^{m-2} = 2^{m+1}$ (because $7 < 8$), we obtain

$$a_{m+1} < 2^{m+1}.$$

Thus the property holds for $m+1$. By strong induction the inequality $a_n < 2^n$ holds for all $n \in \mathbb{N}$. $\square$

Solution 7. Matchstick Game: Two piles start equal (say size $m \geq 1$). Players alternate; on a turn you remove ≥ 1 sticks from exactly one pile. The player who removes the last stick(s) wins. We show Player 2 has a winning strategy.

We prove by *strong induction* on the common pile size m that:

$\mathcal{P}(m)$: From the position (m, m) with Player 1 to move, Player 2 has a winning strategy.

Base case ($m = 1$). The initial position is $(1, 1)$. Whatever Player 1 does, they must remove 1 from one pile, leaving $(0, 1)$. Player 2 then removes the remaining 1 from the other pile and wins. Hence $\mathcal{P}(1)$ holds.

Inductive step. Fix $m \geq 2$ and assume $\mathcal{P}(k)$ holds for all $1 \leq k < m$. Consider the starting position (m, m) with Player 1 to move. Suppose Player 1 removes r sticks from one pile, where $1 \leq r \leq m$, resulting in the position

$$(m - r, m).$$

Player 2 responds by removing exactly r sticks from the other pile (this is legal since the untouched pile has m sticks), producing the position

$$(m - r, m - r).$$

Now it is Player 1's turn on a *smaller equal-piles* position with common size $m - r < m$. By the strong inductive hypothesis $\mathcal{P}(m - r)$, from any position (k, k) with $k < m$, if Player 1 is to move, then Player 2 has a winning strategy. Therefore, after mirroring to $(m - r, m - r)$, Player 2 is in a provably winning situation going forward.

Since Player 2 can mirror any legal move r made by Player 1 and thereby reduce the game to a strictly smaller equal-piles position on which the inductive hypothesis applies, Player 2 can force a win from (m, m) .

Thus $\mathcal{P}(m)$ holds. By strong induction, $\mathcal{P}(m)$ holds for all $m \geq 1$, and Player 2 has a winning strategy from any equal-piles start.

(*Strategy summary.*) Player 2's winning strategy is the **mirror strategy**: after Player 1 removes r from one pile, Player 2 removes r from the other pile, maintaining equality. Eventually the position reaches $(0, 0)$ immediately after Player 2's move, so Player 2 makes the last move and wins. **Solution 8a.** Base $n = 1$: binary 1.

Assume holds up to n , then $n + 1$ obtained by adding 1 to binary expansion, still representation with digits $c_i \in \{0, 1\}$. Thus all n have binary form.

Solution 8b. If n had two distinct binary expansions, subtracting them gives a nonzero integer whose binary expansion sums to zero, impossible. Hence representation unique.

Bijections

Solution 1. (Bijecting $\mathbb{N}$ with $\mathbb{Q}^+$ and then with $\mathbb{Q}$)

Step 1: Enumerate coprime pairs $(p, q) \in \mathbb{N}^2$ (with $\gcd(p, q) = 1$). Arrange $\mathbb{N} \times \mathbb{N}$ in a grid and traverse by anti-diagonals:

$$(1, 1); (1, 2), (2, 1); (1, 3), (2, 2), (3, 1); \dots$$

Formally, for $s = 2, 3, 4, \dots$ list all pairs (p, q) with $p + q = s$ in increasing p . While traversing, *skip* pairs with $\gcd(p, q) \neq 1$. This yields a sequence

$$(p_1, q_1), (p_2, q_2), (p_3, q_3), \dots$$

of coprime pairs. Define $r_n = \frac{p_n}{q_n} \in \mathbb{Q}^+$.

Injectivity onto $\mathbb{Q}^+$. If $r_m = r_n$ then $\frac{p_m}{q_m} = \frac{p_n}{q_n}$ with $\gcd(p_m, q_m) = \gcd(p_n, q_n) = 1$. Uniqueness of reduced form implies $p_m = p_n$ and $q_m = q_n$, hence $m = n$.

Surjectivity onto $\mathbb{Q}^+$. Given any $\frac{a}{b} \in \mathbb{Q}^+$ in lowest terms ($\gcd(a, b) = 1$), the pair (a, b) occurs on the diagonal $a + b$; we never skip it because it is coprime. Thus some n has $(p_n, q_n) = (a, b)$, so $r_n = \frac{a}{b}$.

Therefore $n \mapsto r_n$ is a bijection $\mathbb{N} \rightarrow \mathbb{Q}^+$.

Step 2: Extend to all rationals $\mathbb{Q}$. Define a bijection $\sigma : \mathbb{N} \rightarrow \{0\} \cup \{\pm r_n : n \in \mathbb{N}\}$ by

$$\sigma(1) = 0, \quad \sigma(2n) = -r_n, \quad \sigma(2n+1) = r_n \quad (n \in \mathbb{N}).$$

This is clearly injective, and every element of $\{0\} \cup \{\pm r_n\}$ is hit, so it is bijective. Since $\{0\} \cup \{\pm r_n\} = \mathbb{Q}$, we obtain a bijection $\mathbb{N} \rightarrow \mathbb{Q}$.

Solution 2. (Bijecting $\mathbb{N}$ with $\mathbb{N}^3$)

Define $f : \mathbb{N}^3 \rightarrow \mathbb{N}$ by

$$f(a, b, c) = 2^a 3^b 5^c \quad (a, b, c \in \mathbb{N}).$$

Injectivity. If $f(a, b, c) = f(a', b', c')$ then $2^a 3^b 5^c = 2^{a'} 3^{b'} 5^{c'}$. By the Fundamental Theorem of Arithmetic (unique prime factorization), the exponents on each prime must match, so $a = a'$, $b = b'$, $c = c'$.

Surjectivity onto the image and inverse description. The image consists of positive integers whose prime factors are only 2, 3, 5; the inverse is given by taking the 2-, 3-, and 5-adic exponents:

$$f^{-1}(n) = (v_2(n), v_3(n), v_5(n)),$$

where $v_p(n)$ is the exponent of the prime p in the factorization of n . Thus f is a bijection between $\mathbb{N}^3$ and its image $S = \{2^a 3^b 5^c : a, b, c \in \mathbb{N}\}$.

From S to $\mathbb{N}$. Since S is an infinite subset of $\mathbb{N}$, enumerate S in increasing order (or by any computable bijection) to get a bijection $g : \mathbb{N} \rightarrow S$. Then $g^{-1} \circ f : \mathbb{N}^3 \rightarrow \mathbb{N}$ is a bijection. Hence $\mathbb{N}$ and $\mathbb{N}^3$ are in bijection.

(Alternative direct route.) Because $\mathbb{N} \times \mathbb{N}$ bijects with $\mathbb{N}$ (e.g., Cantor pairing), so does $\mathbb{N}^3 = (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$.

Uncountability of $[0, 1]$

Solution. (Cantor's diagonal argument with non-ambiguous digits)

Assume for contradiction that there is a bijection $L : \mathbb{N} \rightarrow [0, 1]$. Write each $L(n)$ in its (chosen) decimal expansion

$$L(n) = 0.d_{n1}d_{n2}d_{n3}\dots \quad \text{with } d_{nk} \in \{0, 1, \dots, 9\}.$$

To avoid the ambiguity of terminating decimals (e.g. $0.249999\dots = 0.25$), fix the convention that *no number in the list is written with a tail of repeating 9s*. This is always possible: if a number has two expansions, choose the non-9-tail version.

Construct a new number $x = 0.e_1e_2e_3\dots \in [0, 1]$ by the rule

$$e_k = \begin{cases} 5, & \text{if } d_{kk} \neq 5, \\ 6, & \text{if } d_{kk} = 5. \end{cases}$$

Thus for each k , we have $e_k \neq d_{kk}$, and in particular $e_k \in \{5, 6\}$, so x does not end with repeating 9s.

For every $n \in \mathbb{N}$, the number x differs from $L(n)$ in the n -th decimal place: the n -th digit of x is $e_n \neq d_{nn}$, whereas the n -th

digit of $L(n)$ is d_{nn} . Hence $x \neq L(n)$ for all n , contradicting the assumption that L is surjective. Therefore no bijection $\mathbb{N} \rightarrow [0, 1]$ exists; the interval $[0, 1]$ is uncountable.